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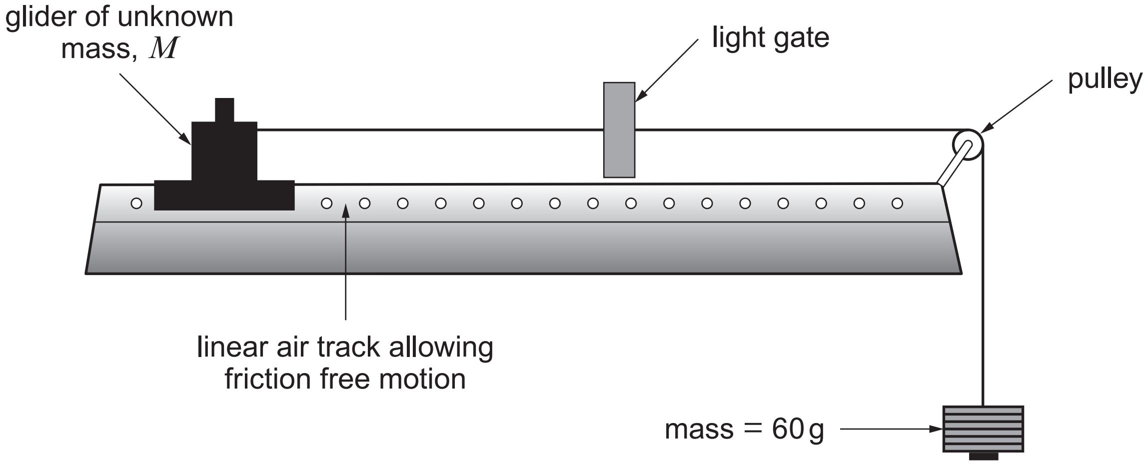
3. Emma uses the following apparatus to determine the unknown mass, M, of a glider.



Emma takes three readings of acceleration using a mass of 60 g. She does this by attempting to hold the glider and releasing it from rest. The following values for acceleration are determined.

Trial	1	2	3
Acceleration / m s^{-2}	0.76	0.71	0.79

(a) (i) Calculate the mean acceleration along with the **absolute** uncertainty in its value. [2]

....

....

....

(ii) Calculate the **percentage** uncertainty in the acceleration. [1]

....

....

(b) The 60 g mass consists of 6 slotted masses, each of 10.0 g ± 0.1 g.

(i) Determine the **percentage** uncertainty in the 60 g mass. [1]

....

....

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(ii) Hence determine the unknown mass, M, of the glider along with the **absolute** uncertainty in its value. Give both values to an appropriate number of significant figures. [5]

(c) Suggest how Emma could reduce the uncertainty in her value for M. [1]

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